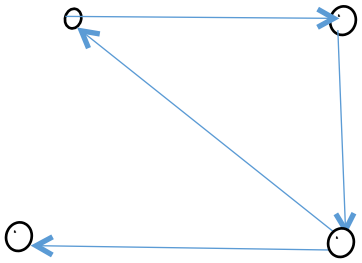
1. Define AVL Trees. Explain its four rotation types . Create AVL Tree for given array 55, 25, 65, 9 ,8 , 15.
2. Define Heap.Construct bottom up heap for the list 2,9,7,6,5,8. Obtain its time complexity.
3. Explain Decision Tree with an example.
4.Design Horspools algorithm for string matching. Apply Horspools algorithm to find the pattern BARBER in the text: JIM_SAW_ME_IN_A_BARBERSHOP
5.Find out warshall's algorithm time complexity.Apply Wrashall's Algorithm to compute transitive closure (path matrix) for given graph below



6. What are Huffman Trees? Construct the Huffman tree for the following

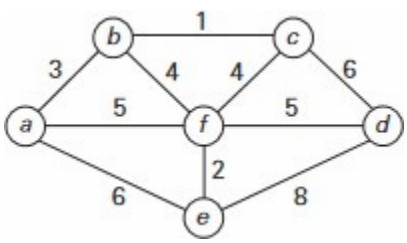
Character	A	B	C	D	E	F
Probability	0.10	0.15	0.50	0.10	0.10	0.05

data.Encode DAD-CBE using Huffman Ecoding.

7. Illustrate N queen's problem using backtracking to solve 4-Queens problem
8) . Explain the following with examples
i) P problem
ii) NP Problem
iii)NP- Complete problem
Iv) NP – Hard Problems
8. Give Branch and Bound Approximation Algorithm.
9. Define heap. Explain the properties of heap along with its representation
10 Define transitive closure of a graph. Apply Warshalls algorithm to compute transitive closure of a directed graph

	a	b	c	d
a	0	1	0	0
b	0	0	0	1
c	0	0	0	0
d	1	0	1	0

10. Construct minimum cost spanning tree using Kruskals algorithm for



11. What is backtracking? Apply backtracking to solve the below instance of sum of 12.subset problem S={5,10,12,13,15,18} d=15

Solve the all pairs shortest path problem for the given adjacency matrix as shown below.

	a	b	c	d
a	0	infinity	3	infinity
b	2	0	infinity	infinity
c	infinity	7	0	1
d	6	infinity	infinity	0

13.Write prims algorithm and give its time complexity.